

Smart Waste Monitoring System

Analytics & Performance Report

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System Overview

This report presents analytics generated from the smart waste monitoring system. The system analyses bin fill levels, identifies collection priorities using a rule-based DSS, and logs collection events separately from sensor telemetry.

Metric	Value
Total Telemetry Records	1
Average Fill Level	35.0%
Critical Records (≥80%)	0
Highest Risk Bin	BIN-001
Highest Fill Level	35%
Total Collection Events	0
Most Collected Bin	N/A
Avg Fill at Collection	0%

Historical Trend Analysis

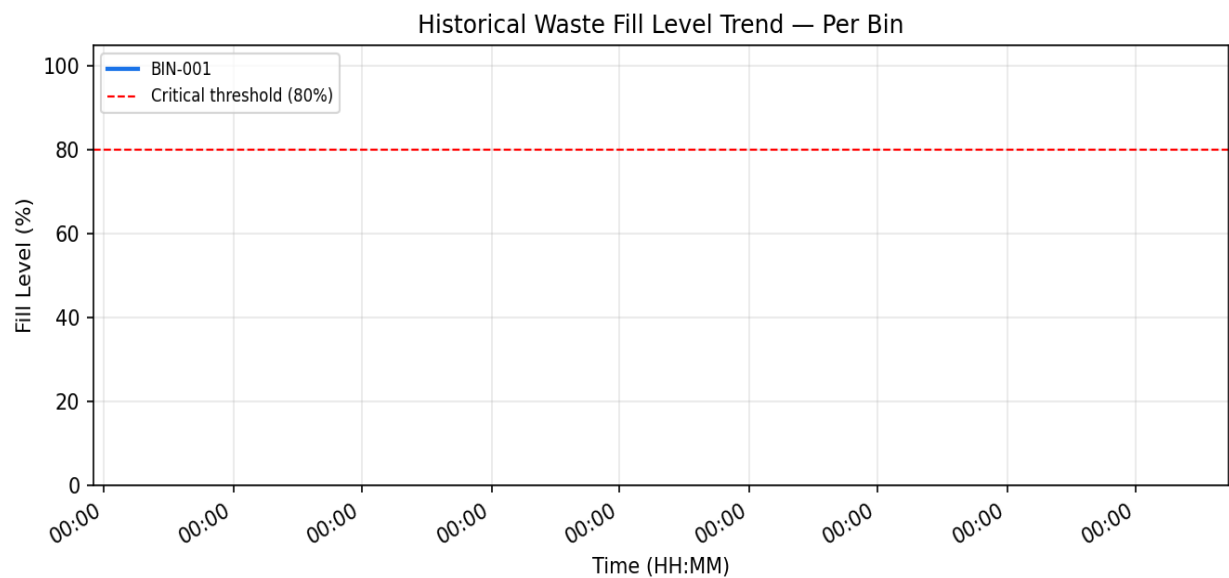


Figure 1. Historical waste fill level trend across recorded sensor readings.

Risk Distribution Analysis

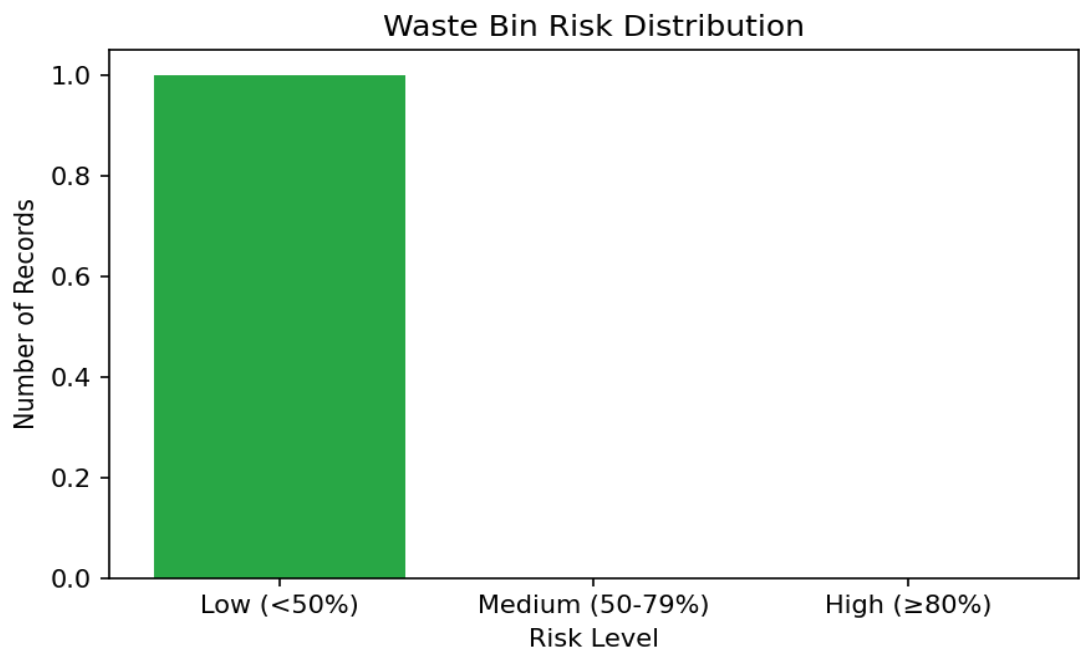


Figure 2. Distribution of low, medium, and high-risk waste records.

Collection Events Summary

The following table summarises bin collection events recorded in the *collection_events* table — stored separately from sensor telemetry to enable clean queryability of actual emptying events.

No collection events have been recorded yet. Events appear here once bins reach the 95% threshold during simulation.

System Assessment

The analytics module successfully evaluates historical waste accumulation trends and identifies high-priority collection requirements. The rule-based DSS applies configurable fill-level thresholds to generate actionable recommendations. Collection events are stored in a dedicated table, enabling independent analysis of collection frequency and bin-level performance.